3 A student drops a heavy ball from a vertical height of 1.8 m above the ground. The ball then falls to the ground. It does **not** bounce after hitting the ground.

(a) Describe the transfers of energy of the ball between stores from when the ball begins to fall to when it reaches the ground.

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..... [3]

(b) Calculate the maximum speed of the ball. Ignore air resistance.

Show your working.

maximum speed = ... [3]

[Total: 6]

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